# Do Now · retrieval
## Signed integers & binary arithmetic

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Choose a type

### Q2

Keep the leading zeros

### Q3

Read an unsigned byte

### Q4

Can it fit?

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Sign magnitude — The leftmost bit stores the sign. The remaining bits store the positive magnitude. — CODE:
10000101
sign = 1
magnitude = 0000101 = 5
value = −5

## Core idea 2

Two’s complement — The most significant bit has a negative place value. — CODE:
11111011
= −128 + 64 + 32 + 16 + 8 + 2 + 1
= −5

## Core idea 3

Add using binary carries — CODE:
0 + 0 = 0
0 + 1 = 1
1 + 1 = 10
1 + 1 + 1 = 11 — Keep the fixed number of result bits. A carry beyond the width is discarded.

## Annotate the key visual

Explain which labels or features show the main process.

# New Learning · applying the ideas

## Core idea 4

Overflow and subtraction — Signed overflow occurs when the mathematical result is outside the representable range. Two same-sign inputs producing an opposite-sign result is a warning sign.

## Vocabulary · use the correct technical terms

Sign magnitude — A signed representation with a sign bit followed by magnitude bits.

Two’s complement — A signed binary representation whose most significant bit has a negative place value.

Magnitude — The size of a value without its sign.

Most significant bit — The leftmost bit in a binary representation.

## Independent skill check · apply the lesson

Using 8-bit two's complement, calculate 23 + 19 and 18 - 29. Show the binary working and signed results.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

Encode −13 in two representations

## Follow the worked method

1. Write +13 in eight bits — CODE:
13 = 8 + 4 + 1
+13 = 00001101 — Both signed representations begin from the positive magnitude.

2. Invert, then add 1 — CODE:
00001101
invert → 11110010
add 1 → 11110011

3. Convert the two’s-complement result to denary — CODE:
−128 + 64 + 32 + 16 + 2 + 1 = −13 — Invert-and-add-one negates a value in a fixed width.

## Why this method works

Identify the decision, rule or invariant that makes the method correct.

# We Do · solo guided practice

Work ALONE. Try first; read the clues one at a time if you get stuck.

Complete the steps below. Check your answer only on page 8.

## YOUR TASK

Build the answer

## CLUES · uncover one at a time

Clue 1: For 8-bit two's complement, the leftmost place has weight -128.

Clue 2: Write the weights of every bit set to 1 in 11100110.

Clue 3: Add those signed place values; the final number should be negative.

## BUILD YOUR ANSWER · in BLACK PEN

Step 1: List the signed place values.

Step 2: Add the signed values.

Step 3: State the denary value and check the sign.

## CHECK BEFORE TURNING THE PAGE

Have you shown every step? The worked self-check is at the END on page 8.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Two representations

Represent −37 in (a) sign magnitude [2] and (b) two’s complement [2]. Show the positive binary value and the change made for each method.

Check that you have used the technical terms and explained your decisions.

# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 2

Explain an overflow
A processor adds 01110000 and 00100000 as eight-bit two’s-complement integers.
State both input values in denary. [1]
Give the stored eight-bit result. [1]
Explain whether signed overflow has occurred. Refer to the mathematical total and the representable range. [2]
State the minimum signed two’s-complement width needed to store the correct result. [1]

Check that you have used the technical terms and explained your decisions.

# Exit Ticket · reflect and retrieve

## Exit Ticket

In 8-bit two’s complement, the signed range is −128 to +127.

## Use key vocabulary from memory

1. Explain Sign magnitude in your own words.

2. Explain Two’s complement in your own words.

3. Explain Magnitude in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

## END SELF-CHECK · only after you have attempted the work

**PAGE 5 · Guided We Do solution:**

11100110 = -128 + 64 + 32 + 4 + 2 = -26.

**PAGE 3 · Independent skill check:**

23 + 19: 00010111 + 00010011 = 00101010 (+42). 18 - 29: 00010010 + 11100011 = 11110101 (-11).

Correct any errors in GREEN PEN. Write one improvement.

YOU DO ANSWERS: rhsccs.org/lessons/h446/lesson-002/answers.html